Jonathan Hummel
Homework 4
RBIF 101

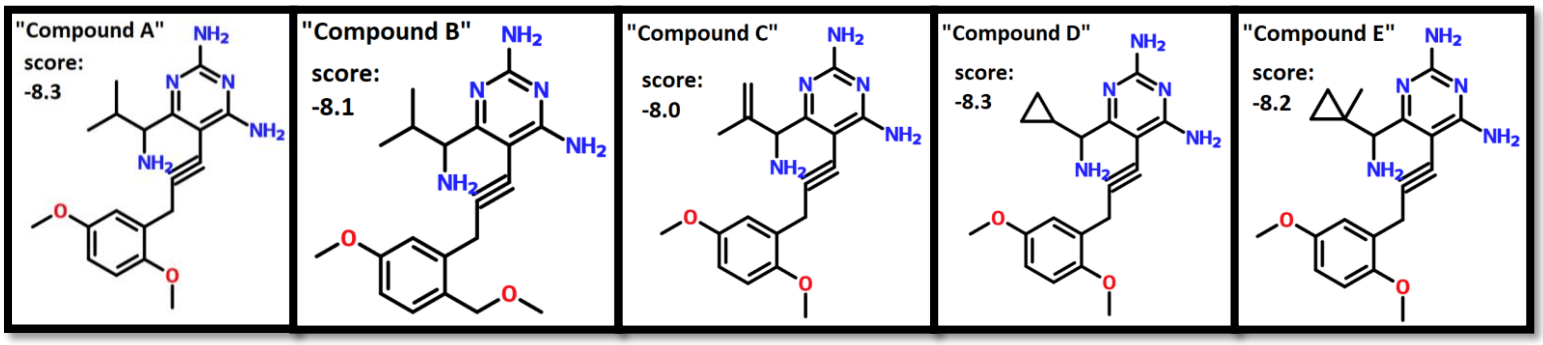
Question A:

My ligand design rationale was to take an iterative approach and attempt to see if I could make incremental improvements to one of the 2 propargyl-linked scaffolds presented in the paper by *Paulsen et al.* [1]. My goal was primarily to improve the selectivity of the inhibitor for CgDHFR and CaDHFR over HuDHFR while hopefully maintaining or improving binding affinity/potency. This goal was inspired by the statements in the paper by *Paulsen et al.* about the differences in loop position between the *Candida* and human DHFR binding sites, and the potential for using this to design a potent and selective compound [1]. Looking at the *Candida* species and human DHFR structures from the paper in PyMol did confirm a less restricted *Candida* binding pocket (discussed further in **Question D**) [1]. I chose Scaffold B as the starting point because when comparing the IC_{50} values from Table 1 by *Paulsen et al.,* it seemed that compounds 4 and 5 showed the best selectivity for the *Candida* species DHFRs over huDHFR [1]. It is worth noting that the IC_{50} values of the Scaffold B compounds were some of the worst for CaDHFR in Table 1, but my focus was on selectivity over the human ortholog, which was still good for CaDHFR [1]. I decided to prioritize *Candida glabrata*, as it poses a newer threat, so I performed docking of my designs against one of the CgDHFR structures from the paper (3QLZ) [1]. That said, during the design process I fetched and considered the other 5 DHFR structures from both *Candida* species (3QLW, 3QLS, 3QLR, 3QLY, and 3QLX), as well as the one for HuDHFR (1KMV) [1]. This enabled me to consider more of the "full picture" when deciding what modifications to try, particularly with regard to selectivity for the *Candida* targets. Because the 1-click docking tool on Mcule.com is so fast and easy to use [2], my design process consisted of this loosely iterative process:

1. Try docking a small modification to compound 4 or 5 (i.e. Scaffold B) by considering all structures and ligands from the paper within PyMol (even those involving Scaffold A) [1]. The structure for CgDHFR complexed with NADPH and compound 5 was the target (3QLZ), and the side-chain carboxylate carbon of the conserved binding site residue **GLU-32** was used as the center [1].
2. Import the docked poses and choose the best-scoring one that was still comparable to the experimentally determined poses from the paper (i.e. still feasible with a nearby NADPH) [1].
3. Consider the docking of the top pose relative to the compounds from the paper, with a particular focus on features that might reduce binding stability with HuDHFR (1KMV) while not negatively affecting binding with the *Candida* species DHFRs [1].
4. Go back to Step 1. Build on the current modification if it was "successful" (i.e. score and apparent selectivity), or try a new modification to compound 4 or 5 if unsuccessful (i.e. failed to properly/feasibly dock and/or relatively bad scores).

I went through over 19 iterations as described above, and the 5 ligands/poses shown on the next page and discussed in **Question B** were the best with regards to docking score, docking feasibility (in the presence of NADPH), pose similarity to ligands from the paper, and features that might diminish HuDHFR binding efficiency. I tried many things that will not be shown or discussed, such as adding R groups to the same position in Scaffold B as the R_1 position in Scaffold A (see Table 1 in the paper) [1]. Some of these may not have been bad ideas, but they did not dock as well as the ones discussed next or did not appear to meet the primary goal of improving potential *Candida* species selectivity.

Below are diagrams of the 5 ligands I chose via the process described on the previous page. For each compound, the reported score is for the highest-scoring pose from 1-click docking [2], which is the pose that will be shown and discussed in **Question B**:



As seen above, compounds that docked to CgDHFR (3QLZ) with the best scores and in a similar fashion to compounds 4 and 5 by *Paulsen et al.* had an amine group attached to the first carbon in the R group [1]. I tried a hydroxyl group in the same position with slightly worse and less reliable docking results. Modifications to the other parts of this group were largely driven by my trying to add structure that fit in the *Candida* species' DHFR binding sites while making the same binding pose less sterically feasible within HuDHFR (discussed later in **Question D**). I tried several modifications to both methoxy groups off of the benzene ring, but most did not dock nearly as well as compound B shown above.

Question B:

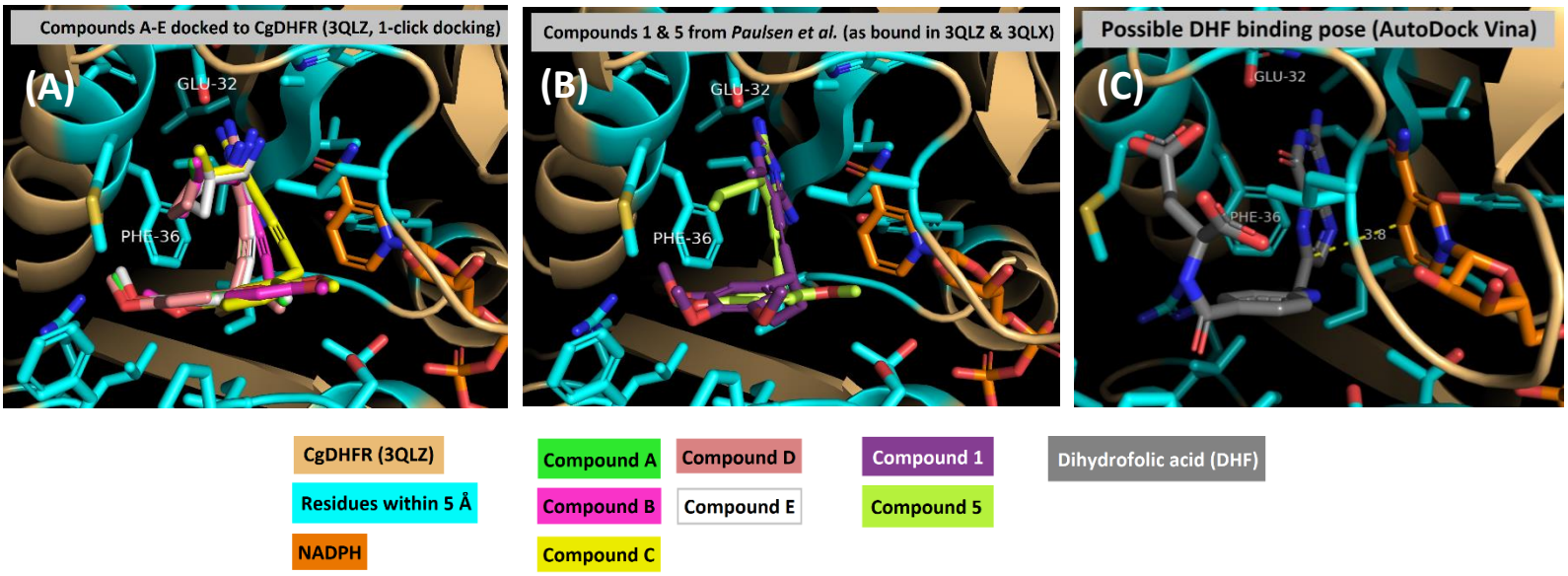
I would say that overall, my rationale seems like it was successful in making incremental improvements to the compounds based on Scaffold B in the paper by *Paulsen et al.* [1]. To get some perspective on how my compounds docked compared to known DHFR ligands, I used the 1-click docking tool to dock several compounds reported to inhibit DHFR of *Candida* species, as well as the substrate dihydrofolate [1,3,4]. **Table 1** on the next page compares the docking scores for those compounds to mine, and compounds are ranked by the score of their highest-scoring *feasible* pose (see **Table 1** footnotes). I know from the Week 7 reading materials that the scoring used to rank docking results does not reliably/perfectly predict which molecules bind most effectively in reality [5,6]. I also know from using the 1-click docking tool that the poses obtained on each run are non-deterministic [2]. Another complicating factor is that the target 3QLZ only captures one conformation of CgDHFR (bound to compound 5 and NADPH) [1]. As such, I am taking the results in **Table 1** with some level of caution, with the understanding that if I had run docking 10 or 100 times for each ligand, the results might paint a different picture. It can be seen in **Table 1** that compounds A, D, and E had better docking scores than all others tested, including the ones from *Paulsen et al.* that they were based on [1]. Compounds B and C were also competitive with most compounds from other sources. Once again, this is not to say that the compounds I designed would perform better in reality than the ones from *Paulsen et al.* or *Chan et al.*, which were shown experimentally to have both promising IC_{50} values (low µM down to nM) and selectivity relative to HuDHFR [1,3]. That said, I do think the results in the table can be taken as encouraging that the ligands I designed may also have potential as inhibitors. Also, this would not be surprising, as I was only making minor changes to a demonstrated scaffold/strategy [1]. With more time, it would have been interesting to recreate **Table 1** on the next page using software like AutoDock Vina and potentially compare higher-quality poses for these compounds.

1-click docking score comparison to known ligands* **Table 1**

Name**	Scaffold	Best Pose***	Source
compound D	propargyl-linked antifolate (Scaffold B)	-8.3	Based on **compounds 4 & 5**
compound A	propargyl-linked antifolate (Scaffold B)	-8.3	Based on **compounds 4 & 5**
compound E	propargyl-linked antifolate (Scaffold B)	-8.2	Based on **compounds 4 & 5**
8d	5-(Arylthio)-2,4-diaminoquinazoline	-8.2	*Chan et al., 1995* (Table 7) [3]
compound B	propargyl-linked antifolate (Scaffold B)	-8.1	Based on **compounds 4 & 5**
compound C	propargyl-linked antifolate (Scaffold B)	-8	Based on **compounds 4 & 5**
8s	5-(Arylthio)-2,4-diaminoquinazoline	-7.9	*Chan et al., 1995* (Table 7) [3]
dihydrofolate		-7.6	Wikipedia [4]
13d	5-(Arylthio)-2,4-diaminoquinazoline	-7.5	*Chan et al., 1995* (Table 8) [3]
compound 1	propargyl-linked antifolate (Scaffold A)	-7.4	*Paulsen et al., 2011* (Table 1) [1]
compound 4	propargyl-linked antifolate (Scaffold B)	-7.2	*Paulsen et al., 2011* (Table 1) [1]
compound 5	propargyl-linked antifolate (Scaffold B)	-7.1	*Paulsen et al., 2011* (Table 1) [1]

* Docking to CgDHFR (3QLZ) with GLU-32's side-chain carboxylate carbon as binding site center
** Compound 2 from *Paulsen et al.* was docked, but all poses clashed notably with NADPH
*** Score for the highest-scoring pose(s) that did not conflict with NADPH presence and/or were generally consistent with binding seen in crystal structures from *Paulsen et al.* (for compounds 1, 2, 4, and 5 or those inspired by them)

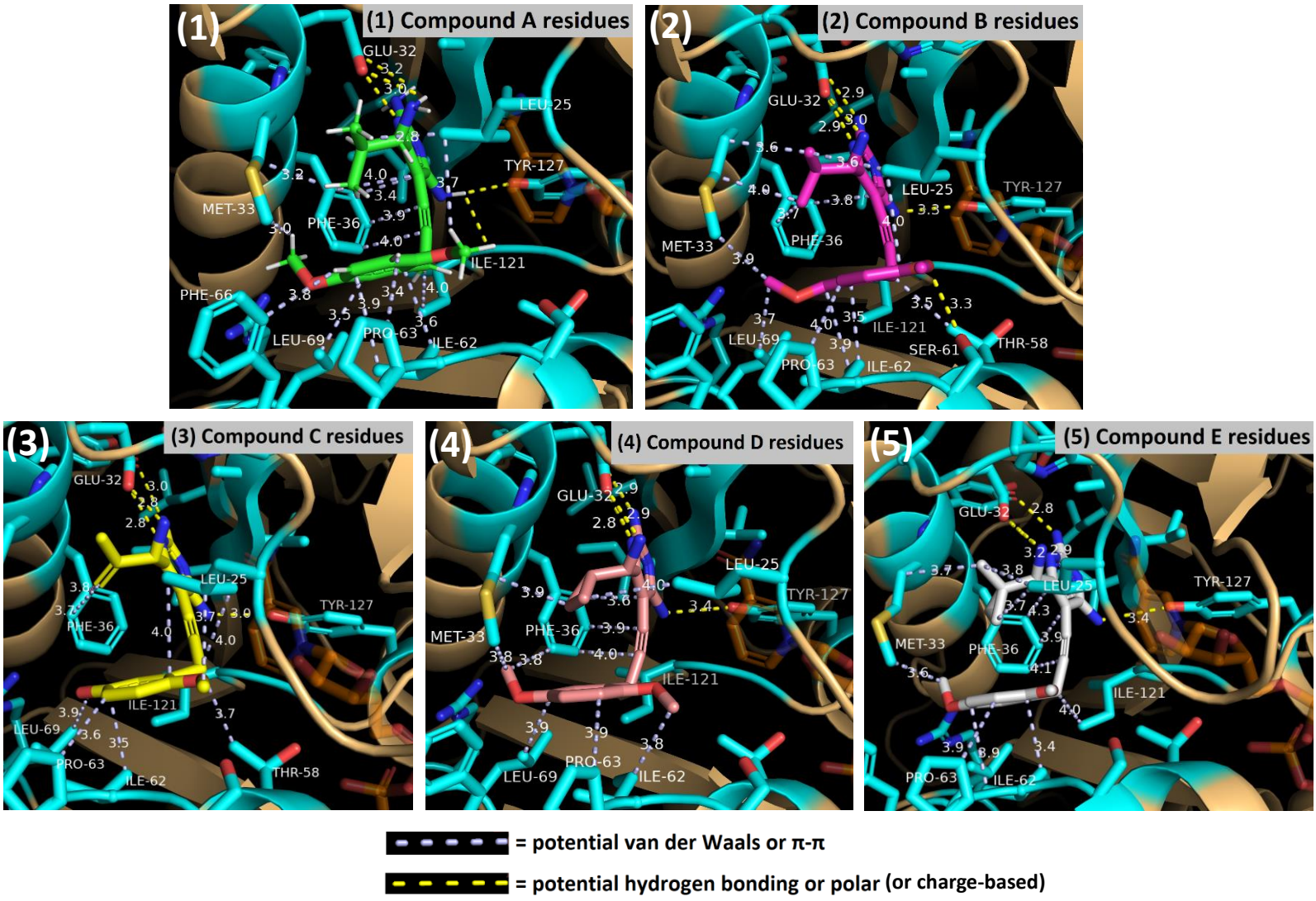
Of course, more interesting than the scoring are the poses themselves. The figures below compare the best 1-clicking docking poses for compounds A-E (**Figure A**) to the poses for compounds 1 and 5 (**Figure B**) from the crystal structures in the paper (3QLZ and 3QLX) [1]. Central binding site residues **GLU-32** and **PHE-36** are included for reference in all images, but key residue interactions will be shown/discussed in more detail later. Also included below is what I think to be a *roughly* feasible pose for dihydrofolic acid (DHF) during catalysis of its reduction via NADPH based on a paper I found about this reaction (**Figure C**) [7]. The distance measurement roughly indicates the positioning of key atoms in the reaction [7], and the pose was generated by AutoDock Vina (via DockingPie, i.e. <u>not</u> the pose from 1-click docking reported in **Table 1**). This DHF pose seems consistent with crystal structures of DHF-DHFR complexes from *E. coli* (1RF7) and *P. falciparum* (4DPD), at least with regards to the double-ring and the approximate positioning of the to-be-reduced carbon close to the relevant part of NADPH [7-9].



As seen above in **Figure A**, the compounds I designed based on compounds 4 and 5 from the paper docked in similar poses to each other [1]. By comparing **Figures A & B**, it is clear that compounds A-E docked with the CgDHFR structure comparably to the binding seen for the propargyl-linked antifolates in the crystal structures from *Paulsen et al.* [1]. It is worth noting that the scores in **Table 1** for compounds 1 and 5 were for similar but not identical poses to the actual ones crystalized by *Paulsen et al.* and shown in **Figure B** above [1]. A possible

binding pose for the substrate DHF is shown in **Figure C** and illustrates geometric and chemical properties that likely enable it to bind this site during catalysis of its reduction via NADPH. The measurement shows that the key atoms are brought within proximity even in this very approximate pose [7]. All inhibitors shown in **Figures A & B** share at least some structural and chemical similarity to DHF that enable them to bind to the same site. Based on these figures, it seems that my design strategy was a success in that the new compounds dock similarly and plausibly relative to known ligands. Once again, the modifications made were only incremental relative to compounds 1-5 from *Paulsen et al.* [1], so this is not too surprising.

Which residues are most important for the binding of the compounds A-E to CgDHFR? Figures 1-5 below show the best feasible 1-click docking poses for compounds A-E docked to CgDHFR (3QLZ), with possible interactions with key residues noted by measurements. The yellow dashed lines indicate polar interactions (or related), while the greyish-blue lines indicate van der Waals interactions (or related). The proposed van der Waals interactions are far from exhaustive and are only meant to provide an idea of the closest residues for such interactions in each pose. NADPH is shown in orange for reference, but it is partially transparent.



= potential van der Waals or π-π

= potential hydrogen bonding or polar (or charge-based)

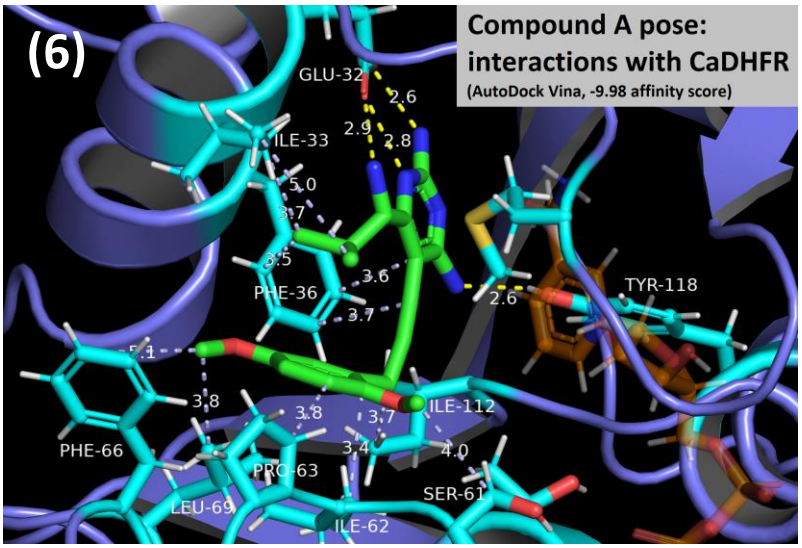
As seen in **Figures 1-5** above, interactions with **GLU-32**'s negatively charged (at physiological pH) carboxylate group are probably central to the binding of compounds A-E to CgDHFR [10]. This residue is close enough to two

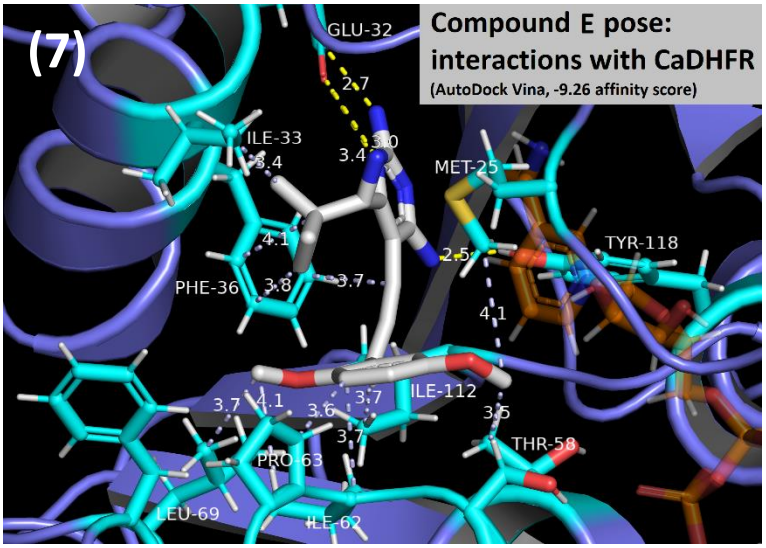
amine groups, which could be anywhere from partially to fully positively charged, so polar or even full-charge interactions are possible. Interactions also seem possible with the pyrimidine nitrogen nearest to **GLU-32**, I think due to the delocalized nature of electrons shared in the ring (causing N states with partial positive charges). Another polar/charge-based interaction common to all poses/compounds could happen between the partially positive hydrogen of **TYR-127**'s hydroxyl group and the lone pair electrons of an amine nitrogen attached to the pyrimidine ring. While the distances involved are a little on the high end for hydrogen bonding, this binding pocket is quite hydrophobic overall, so imagine this interaction is strengthened. The same goes for the polar/charge-based interactions mentioned earlier with **GLU-32**. It is worth noting that PyMol detected a hydrogen bond in some poses between an amine hydrogen and the lone pair(s) of **ILE-121**'s backbone oxygen. This is only explicitly shown in **Figure 1** for compound A. While the polar/charge interactions form key anchor points for the ligands, many hydrophobic side chains within proximity likely have a notable cumulative effect that stabilizes these compounds within the binding pocket. The specific residues involved appear to change a little bit depending on the binding pose, but most of the time **PHE-36**, **MET-33**, **LEU-69**, **PRO-63**, **ILE-62**, and/or **LEU-25** appear to be close enough to engage in stabilizing van der Waals interactions with these compounds (see **Figures 1-5**). In some poses the molecule is more to the "right" side of the binding site (as shown in the figures), and it likely depends more on interactions with residues on that side, e.g. **PHE-66** and **MET-33**. In other poses the compound is shifted to the "left" and likely has van der Waals interactions with **THR-58**, and sometimes even a polar interaction with the hydroxyl hydrogen of **SER-61** (ligand's oxygen lone pairs as acceptors). Sometimes the "elbow" at the bottom of the linker is close enough to **ILE-121** for van der Waals (see **Figures 1 & 5**). These are just the nearest interactions that caught my eye, but the *combined* effect of van der Waals with most if not all of these hydrophobic side chains is likely important whichever pose one of these compounds adopts. Essentially, the benzene ring of Scaffold B provides a central platform for van der Waals interactions with the hydrophobic "floor" of the pocket, while the propargyl linker region provides opportunities for such interactions further into the pocket. I imagine as this molecule binds, the pyrimidine ring may even engage in stronger π-π interactions with **PHE-36'**s ring before the polar/charge interactions with **GLU-32** discussed earlier pull it further into the pocket. Unsurprisingly, all of these core structural features of the propargyl-linked scaffolds proposed by *Paulsen et al.* resemble those likely central to DHF's positioning within this same pocket (see **Figure C**) [1,7-9]. The central role of interactions with **GLU-32** and **PHE-36** described above is consistent with observations by *Paulsen et al.* [1]. The same can be said about the importance of certain nearby hydrophobic residues, e.g. **PRO-63**, **LEU-69**, **ILE/MET-33**, **SER-61**, and **ILE- 62** [1].

Question C:

While good docking does not necessarily translate to effective inhibition, I think the details shown and discussed in **Question B** support the possibility that compounds A-E might effectively inhibit CgDHFR. That said, would any of them inhibit <u>CaDHFR</u> just as effectively? This seems likely at face value because the compounds I proposed do not deviate far from those already demonstrated by *Paulsen et al.* to inhibit DHFR in both *Candida* species [1]. Of the 12 residues that were noted to facilitate the binding of compounds A-E to CgDHFR in **Question B**, 10 were identical at homologous 3D positions within a structural alignment between CgDHFR (3QLZ) and CaDHFR (3QLW) in PyMol (not shown). The 2 non-identical residues in CaDHFR were still chemically similar, i.e. hydrophobic. A methionine replaces a leucine at position 25, and an isoleucine replaces a methionine at position 33 (these were aligned in 3D space). The LEU-MET substitution even conserves side chain size, but the MET-ILE substitution does result in a notably more open "left side" of the pocket in CaDHFR compared to CgDHFR (compare **Figures 1 and 6**). Most importantly, however, the key **GLU-32** residue is conserved in both species, as is **PHE-36**. Due to all of these details, I think it is fairly likely that compounds A-E could effectively inhibit both CaDHFR and CgDHFR. To investigate these claims, I decided to dock compounds A and E to a structure for CaDHFR from the paper *Paulsen*

et al. (3QLW) [1]. I ran out of weekly docking attempts on 1-click docking, so I used AutoDock Vina in PyMol via DockingPie. Structures for compounds A and E were extracted from the earlier docking results and prepared by Vina (e.g. adding hydrogens), and 3QLW was also fetched and prepared using Vina in DockingPie. I had AutoDock Vina find 5 poses for each compound, and no residues were set as flexible (though this may have improved the results). The center of the binding site was set as the position of one of the central linker carbons in compound 4 as it appears in 3QLW. **Figures 6 & 7** below show the best scoring pose obtained from AutoDock Vina, as well as plausible interactions with key residues of CaDHFR. It is worth noting that the top 3 poses for compound A had affinity scores of -9.98, -9.67, and -8.08 and were all comparable to that of compound 4 as it appears in 3QLW. The top 2 compound E poses were also similar to the crystalized pose of compound 4 and had affinity scores of -9.26 and -9.21. Ligand hydrogens were present during docking, although they are not all shown below:





= potential van der Waals or π-π

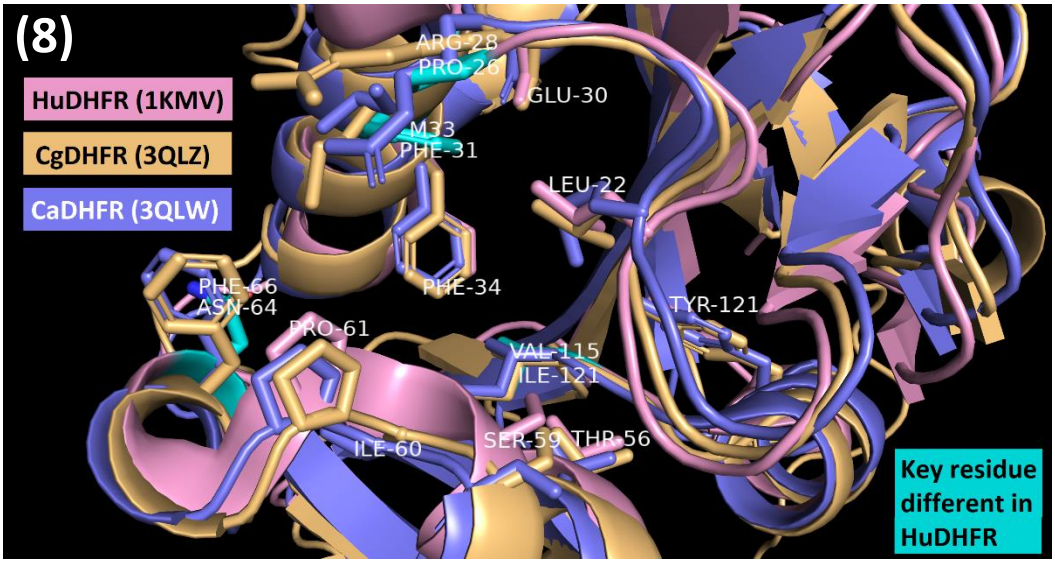
= potential hydrogen bonding or polar (or charge-based)

As seen above, compounds A & E docked with the CaDHFR structure (3QLW) comparably to how they docked with a structure from *C. glabrata* earlier (in **Figures 1-5**). As predicted earlier, **MET-25** and **ILE-33** interact similarly with these ligands to the other hydrophobic residues present at these positions in CgDHFR. Similarly, the key residues and interactions that appeared to facilitate binding in CgDHFR are otherwise identical above in CaDHFR. Based on the similarity of compounds B-D to the ones docked and shown above, I expect they would also be able to dock (and possibly bind) with CaDHFR in a similar fashion. Once again, the poses shown above are also quite similar to how compound 4 crystalized within the binding site of CaDHFR in 3QLW (not shown) [1]. Out of curiosity, I docked compound 4 with the same CaDHFR structure using AutoDock Vina, and the two best poses, one of which was nearly identical to the crystal structure, had affinity scores of -9.31 and -9.29 (not shown) [1]. Despite all the caveats discussed earlier regarding docking scores, the fact that the best AutoDock Vina scores above for compounds A and E were comparable to or better than those obtained for compound 4 does provide a promising reference point. Much like the comparisons made earlier in **Table 1** and **Figures A & B**, this is encouraging because compound 4 was shown to inhibit both CaDHFR and CgDHFR in the paper [1].

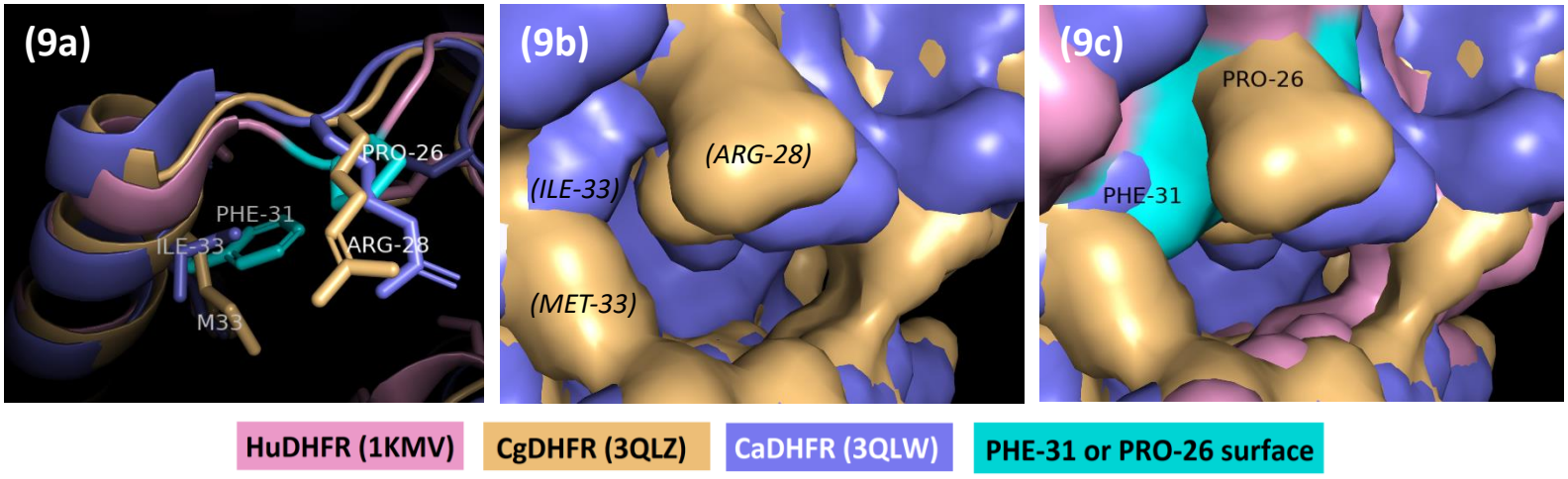
Question D *starts on the next page…*

Question D:

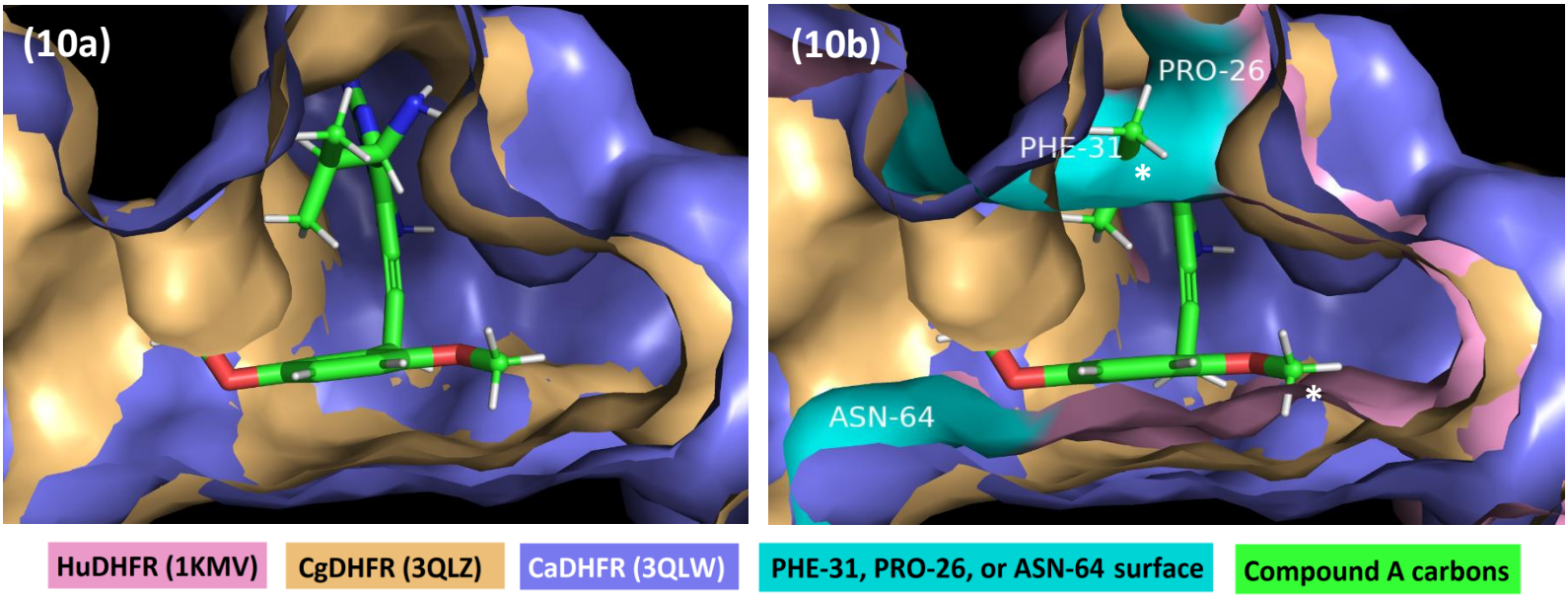
Figure 8 below was prepared in PyMol by fetching structures for HuDHFR (1KMV), CgDHFR (3QLZ), and CaDHFR (3QLW) discussed in the paper by *Paulsen et al.* and using *cealign* to align the *Candida* structures to the human one [1]. Most of the residues involved in inhibitor binding (discussed in **Questions B and C**) are shown as sticks, and the structurally homologous residues in HuDHFR are labeled and shown as sticks. Wherever there was a difference in the human residue at one of these 3D positions, the human residue is colored cyan and the label of the homologous *C. glabrata* residue is also shown directly above or below the human one.



As seen above, many of the residues discussed so far that are important for the propargyl-linked antifolate inhibitors to bind this site are conserved in HuDHFR. Most notably **GLU-30** is in a comparable position to **GLU-32** in the *Candida* proteins, and the same goes for **TYR-121** (compared to **TYR-118/TYR-127** shown and discussed earlier). **PHE-34** in HuDHFR is structurally conserved when compared to **PHE-36** in the *Candida* species' DHFRs. Also, most of the hydrophobic residues in this binding pocket discussed earlier are conserved in HuDHFR. That said, there are a few different residues in the human enzyme. For example, **ASN-64** replaces PHE-66, **VAL-115** replaces ILE-121/ILE-112, **PHE-31** replaces MET-33/ILE-33, and **PRO-26** replaces ARG-28. Some of these residues may provide opportunities to design a more selective inhibitor targeting DHFR of the *Candida* species. Most notably, the shape of the binding pocket is more restricted in HuDHFR due to differences in the positioning of the secondary structures above and below the pocket (i.e. the loop "above" and the loop/small helices "below", as seen in **Figure 8)** [1]. For the top of the human pocket, these spatial differences appear to be at least partly driven by **PHE-31** and **PRO-26**. **PRO-26** is much less flexible than ARG-28 and appears to be involved in bending its loop "downward" via the kink it introduces (see **Figure 9a**). Meanwhile, PHE-31 is larger than MET-33 or ILE-33 and has a different geometric profile. Together these residues appear to create a much more spatially restricted top section of the HuDHFR pocket compared to that of the *Candida* species. This is illustrated in **Figure 9** below:



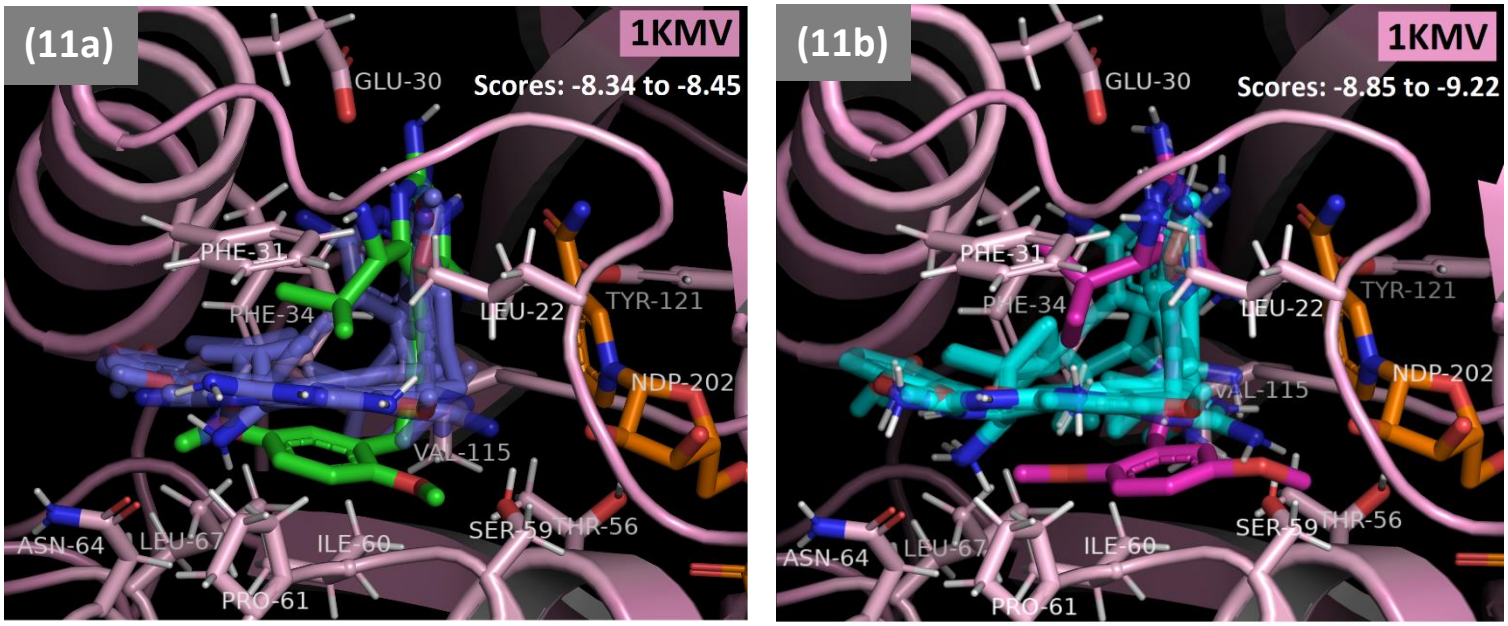
While not the intent of **Figure 9**, it can also be seen in **Figure 9c** on the previous page that the bottom of the binding pocket of HuDHFR is also more restricted (bits of pink surface poking through). Trying to leverage the steric restrictions at the top and bottom of the HuDHFR pocket was actually what motivated the chemical groups present in compounds A-E, designed and shown in **Question A**. I was trying to add groups that, in addition to having decent polar and/or van der Waals interactions with the CgDHFR pocket residues, also had a geometry that might sterically clash with PHE-31 in HuDHFR (hopefully without destabilizing binding to CgDHFR or CaDHFR too much). I figured that such an R group, coupled with the length and straightness of the propargyl linker that likely forces the benzene ring and its attached groups closer to the more restricted "floor" of the HuDHFR pocket, could reduce the stability of an inhibitor specifically in HuDHFR. This rationale is illustrated in **Figure 10** below, which shows compound A docked to CgDHFR with aligned structures from HuDHFR and CaDHFR. **Figure 10** shows a cross-section of the estimated binding pocket surfaces of all three enzymes, with the same compound A docking pose shown earlier in **Figure 1** (i.e. best 1-click docking pose).



By comparing **Figure 10b** to **10a**, it is clear that in the best pose found earlier for CgDHFR, one of the methyl groups in compound A protrudes into the surface of where **PHE-31** would be in the HuDHFR structure (clash marked by an asterisk). As discussed earlier, the surfaces of **PHE-31** and **PRO-26** (partially outside of the foreground of **Figure 10b**) together create a significantly more restricted binding pocket in HuDHFR. The end of one of the methoxy groups off of compound A's benzene ring also comes way too close to **SER-59** at the bottom of the pocket (marked by an asterisk, serine not labeled). Much of that part of compound A is likely too close to HuDHFR residues. **ASN-64** is highlighted mainly for reference and comparison to earlier figures, but it might play a role in the more restricted positioning of the secondary structures at the "bottom" of the pocket. Regardless of the cause, it can be seen in **Figure 10b** that the bottom of the HuDHFR pocket is raised quite a bit compared to the *Candida* orthologs. I am aware that these crystal structures represent only one state of each ortholog, and potential flexibility in the loops around the binding pocket, as well as other degrees of freedom in the target or ligand, mean that my rationale for *Candida* selectivity in a DHFR inhibitor may not work. Perhaps HuDHFR would simply adopt a different conformation to bind compound A similarly to the poses shown for CgDHFR and CaDHFR so far, contrary to what is shown in **Figure 10b**. That said, the compounds I designed are based on compounds demonstrated by *Paulsen et al.* to have notable selectivity for CaDHFR and CgDHFR over HuDHFR [1]. It seems reasonable to think that this selectivity may stem from the straightness and length of the propargyl linker, making the other parts of compounds not as easily accommodated by the HuDHFR binding pocket [1]. Following the same reasoning, I think the modifications I made to Scaffold B from the paper might not be as easily accommodated by HuDHFR as they would be by CgDHFR and CaDHFR. Regardless of the specific approach taken,

I think making full use of the more open geometry of the binding pockets of CaDHFR and CgDHFR (relative to that of HuDHFR) appears to be one of the more viable strategies for improved selectivity.

While far from definitive, I wanted to see if my rationale described above for improving selectivity at least resulted in the inability of compounds A and E to dock with the HuDHFR structure in a pose like the ones captured in the crystal structures presented by *Paulsen et al.*, or those shown earlier for compounds A-E [1]. I used AutoDock Vina to dock compounds A and E to the HuDHFR structure (1KMV, same process described in **Question C**). **Figure 11** below shows the 5 docking poses for each compound, with the highest scoring pose shown opaque and the other poses partially transparent (and score ranges reported). For each compound, its orientation/pose shown earlier when docking to CgDHFR (3QLZ) is also included (i.e. the complex from 1-click docking was aligned to 1KMV and only the compound was shown). This was included to roughly compared how these compounds docked with HuDHFR to how they did with CgDHFR (and effectively CaDHFR).



Compound A, as docked to CgDHFR (reference only)

Compound A poses w/ HuDHFR (AutoDock Vina)
(highest scoring pose not transparent)

Compound E, as docked to CgDHFR (reference only)

Compound E poses w/ HuDHFR (AutoDock Vina)
(highest scoring pose not transparent)

As seen in **Figure 11** above, it does appear that compounds A and E are unlikely to dock with HuDHFR in the same way they did with the target *Candida* orthologs due to the more restricted binding pocket and the shape of the molecules. For each compound, all but one or two poses do not put the pyrimidine and its nearby amine groups anywhere near the conserved **GLU-30** residue for strong polar/charge interactions. The poses that did were not among the highest scoring poses, and the relevant groups were not as close to glutamic acid's side chain as in CgDHFR. The same can be said for hydrogen bonding with **TYR-121**, as it seems that in most poses there is no group close enough to interact. Even for poses with a nearby nitrogen, it is at a more marginal distance to tyrosine. So, the strategy of leveraging a more sterically restrictive HuDHFR binding pocket may have "worked", at least to prevent these inhibitors from engaging in the same important polar/charge interactions that were apparently central to their binding with CgDHFR and CaDHFR. That said, there is still a myriad of van der Waals interactions that these compounds can engage in with HuDHFR residues, as well as some polar ones with **SER-59** and **ASN-64** (depending on the pose). The poses for compounds A and E are quite similar, so I think the notable jump in affinity scores for compound E is due to the ability of its R-group cyclopropane and methyl

components to fill the space better and engage in more van der Waals interactions, particularly with **PHE-34** (but also PHE-31, and other hydroponic residues depending on the pose). With the caveat that affinity scoring is far from definitive, it appears that compound A may be the more selective of the two investigated in **Figure 11**. The score for the best AutoDock Vina pose of compound A in CaDHFR (**Figure 6**) was -9.98 for a comparable pose to the paper [1]. Meanwhile, the affinity score in **Figure 11a** with HuDHFR was -8.45 for a pose not comparable to the paper [1]. The top docking scores for compound E with CaDHFR and HuDHFR were similar, with HuDHFR perhaps having a slightly higher average, although its poses were also not like those demonstrated in the paper [1]. That said, I did not perform docking with AutoDock Vina for a CgDHFR structure for direct comparison to these results (or Vina docking with compounds 1-5, to see if any selectivity improvement might have been gained). Also, all of the Vina affinity scores for these compounds are closer between HuDHFR and CaDHFR than I was hoping. Ultimately, only activity assays and other testing could reveal whether the compounds I designed from Scaffold B selectively bind CaDHFR and CgDHFR over HuDHFR in a meaningfully improved way [1]. Even if the general strategy for selectivity discussed above is sound, the way to most confidently evaluate leads is still experimentation.

References:

[1] Paulsen, J. L., Bendel, S. D., & Anderson, A. C. (2011). Crystal Structures of Candida albicans Dihydrofolate Reductase Bound to Propargyl-Linked Antifolates Reveal the Flexibility of Active Site Loop Residues Critical for Ligand Potency and Selectivity. In Chemical Biology & Drug Design (Vol. 78, Issue 4, pp. 505–512). Wiley. https://doi.org/10.1111/j.1747-0285.2011.01169.x

[2] https://mcule.com/apps/1-click-docking/

[3] Chan, J. H., Hong, J. S., Kuyper, L. F., Baccanari, D. P., Joyner, S. S., Tansik, R. L., Boytos, C. M., & Rudolph, S. K. (1995). Selective Inhibitors of Candida albicans Dihydrofolate Reductase: Activity and Selectivity of 5-(Arylthio)-2,4-diaminoquinazolines. In Journal of Medicinal Chemistry (Vol. 38, Issue 18, pp. 3608–3616). American Chemical Society (ACS). https://doi.org/10.1021/jm00018a021

[4] https://en.wikipedia.org/wiki/Dihydrofolate_reductase

[5] Irwin, J. J., & Shoichet, B. K. (2016). Docking Screens for Novel Ligands Conferring New Biology. In Journal of Medicinal Chemistry (Vol. 59, Issue 9, pp. 4103–4120). American Chemical Society (ACS). https://doi.org/10.1021/acs.jmedchem.5b02008

[6] Kitchen, D. B., Decornez, H., Furr, J. R., & Bajorath, J. (2004). Docking and scoring in virtual screening for drug discovery: methods and applications. In Nature Reviews Drug Discovery (Vol. 3, Issue 11, pp. 935–949). Springer Science and Business Media LLC. https://doi.org/10.1038/nrd1549

[7] Kohen, A. (2015). Dihydrofolate reductase as a model for studies of enzyme dynamics and catalysis. In F1000Research (Vol. 4, p. 1464). F1000 Research Ltd. https://doi.org/10.12688/f1000research.6968.1

[8] Sawaya, M. R. (1997). STRUCTURE OF DIHYDROFOLATE REDUCTASE COMPLEXED WITH DIHYDROFOLATE. Worldwide Protein Data Bank. https://doi.org/10.2210/pdb1rf7/pdb

[9] Yuthavong, Y., Vilaivan, T., Kamchonwongpaisan, S., Charman, S. A., McLennan, D. N., White, K. L., Vivas, L., Bongard, E., Chitnumsub, P., Tarnchompoo, B., Thongphanchang, C., Taweechai, S., Vanichtanakul, J., Arwon, U., Fantauzzi, P., Yuvaniyama, J., Charman, W. N., & Matthews, D. (2012). WILD TYPE PLASMODIUM FALCIPARUM DIHYDROFOLATE REDUCTASE-THYMIDYLATE SYNTHASE (PfDHFR-TS), DHF COMPLEX, NADP+, dUMP. Worldwide Protein Data Bank. https://doi.org/10.2210/pdb4dpd/pdb

[10] https://en.wikipedia.org/wiki/Amino_acid